Cross Product

I. Definition

Vector Operation 3: Cross Product

Let $\mathbf{a} = [a_1, a_2, a_3]$, $\mathbf{b} = [b_1, b_2, b_3]$, and $\mathbf{n}$ is a unit vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

Then

$$\mathbf{a} \times \mathbf{b} = (|\mathbf{a}|\,|\mathbf{b}| \sin\theta)\,\mathbf{n}$$

and so

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}| \sin\theta$$

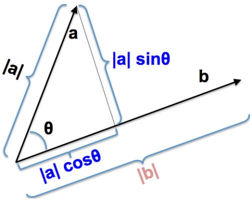
Notes:

- $\mathbf{a}, \mathbf{b} \in \mathbf{R}^3 \quad \Longrightarrow \quad \mathbf{a} \times \mathbf{b} \in \mathbf{R}^3$ (*vector*) and $\mathbf{a}, \mathbf{b} \perp \mathbf{a} \times \mathbf{b}$
- $\mathbf{a} \times \mathbf{b} \neq \mathbf{b} \times \mathbf{a}$ (*not commutative*)
- A way to remember how to calculate a vector product is to use determinants:

$$\begin{aligned}\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \\ &= (a_2 b_3 - a_3 b_2)\mathbf{i} - (a_1 b_3 - a_3 b_1)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k} \\ &= [a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1]\end{aligned}$$

II. Additional Concepts

Geometric Interpretation 1:



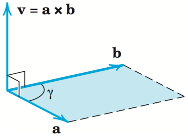
Dot Product:

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= (\text{component of } \mathbf{a} \text{ along } \mathbf{b})(\text{length of } \mathbf{b}) \\ &= |a|\,|b| \cos\theta \end{aligned}$$

Cross Product:

$$\begin{aligned} |\mathbf{a} \times \mathbf{b}| &= (\text{component of } \mathbf{a} \text{ perpendicular to } \mathbf{b})(\text{length of } \mathbf{b}) \\ &= |a|\,|b| \sin\theta \end{aligned}$$

Geometric Interpretation 2:



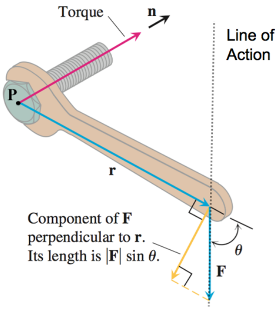
$|\mathbf{a} \times \mathbf{b}|$ = Area of the parallelogram between **a** and **b**

Properties: Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbf{R}^3$ and $d \in \mathbf{R}$ (*a scalar*)

- $d(\mathbf{a} \times \mathbf{b}) = d\mathbf{a} \times \mathbf{b} = \mathbf{a} \times d\mathbf{b}$
- $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
- $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$

III. Application

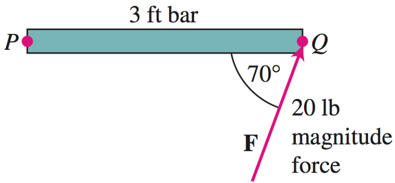
Application: Torque and Moment Force about an axis through point P.



Torque (Moment Force) is the tendency of a force to rotate an object about an axis.

$$\textit{Torque/Moment Force} = \mathbf{F} \times \mathbf{r}$$

Verify that the torque vector in the diagram is pointing in the correct direction.



Ex: Find the Torque, τ, on an axis through P in the bottom diagram.

Soln:
$|\tau| = |\mathbf{F} \times \overline{PQ}| = |\mathbf{F}||\overline{PQ}| \sin 70 \approx (20\,\text{lbs})(3\,\text{ft})(0.94) = 56.4$ ft-lb.

Since our problem is within a plane (say the xy−plane) then the unit normal vector perpendicular to $\mathbf{F}$ and $\overline{PQ}$ is $\mathbf{n} = [0, 0, -1]$ (by right-hand rule)

Therefore, $\tau = |\tau|\mathbf{n} \approx [0, 0, -56.4]$